Today
Boundary Value Problems
Shooting Method
Finite Difference

Announcements
HW5 due Thursday
(no late penalty)

## So far: initial value problems

$$\vec{x} = \begin{bmatrix} x \\ \dot{x} \end{bmatrix} \qquad \dot{\vec{x}} = f(t, \vec{x}) \qquad \begin{matrix} x(0) = a \\ \dot{x}(0) = b \end{matrix}$$ ← initial conditions

## Boundary-value problems

Example: $x(0) = a$ (initial pos.) $\qquad x(T) = b$ (final pos.)

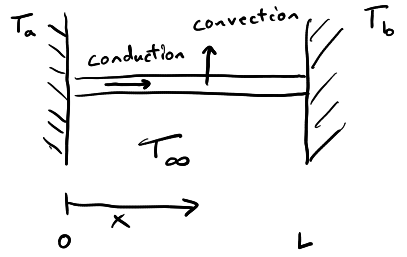
$\dot{x}(0), \dot{x}(T)$ unknown

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$x$ known: Dirichlet B.C. $\qquad$ $\dot{x}$ known: Neumann B.C.

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Example: heated rod




$$\frac{d^2T}{dx^2} + h'(T_\infty - T) + \underbrace{\sigma'(T_\infty - T)}_{\text{radiation}} = 0 \qquad \begin{matrix} T(0) = T_a \\ T(L) = T_b \end{matrix}$$

Solve analytically

$$\vec{y} = \begin{bmatrix} T \\ \frac{dT}{dx} \end{bmatrix} \qquad \frac{d\vec{y}}{dx} = f(t, \vec{y})$$

$$\vec{y}(0) = \begin{bmatrix} T(0) \\ \frac{dT}{dx}(0) \end{bmatrix} = \begin{bmatrix} T_a \\ \frac{T_b - T_a}{L} \end{bmatrix} \quad \text{start with } \frac{T_b - T_a}{L}\text{, refine}$$

ode45(f, [0, L], $\vec{y}(0)$)

## Shooting Method

Start with known IC's, guess unknown ICs
Adjust unknown IC's and "shoot" until we satisfy all BCs

How do we shoot in a smart way?

Let $z \equiv \frac{dT}{dx}$ $\qquad z(0)$ is unknown IC

define $g(T(0), z(0))$ integrates (using RK4/ode45) and returns $T(L)$

$g(T_a, z(0)) = T_b \Rightarrow 0 = g(T_a, \underbrace{z(0)}_{\text{unknown}}) - T_b$ $\qquad f(z(0)) = 0$ nonlinear scalar equation

solve for $z(0)$ using ——

↑ your favorite nonlinear solver
Newton Raphson, Incremental Search, Secant Bisection

can diverge for complex dynamics

# Finite Difference Method

1) Discretize problem into $n$ $\Delta x$ segments
2) Replace derivatives with finite difference (central finite difference)
3) solve resulting equations

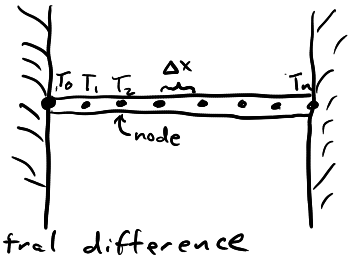
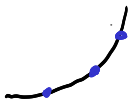



Fig 21.5: central difference

| First Derivative | Error |
|---|---|
| $f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h}$ | $O(h^2)$ |
| $f'(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2})}{12h}$ | $O(h^4)$ |
| **Second Derivative** | |
| $f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2}$ | $O(h^2)$ |
| $f''(x_i) = \frac{-f(x_{i+2}) + 16f(x_{i+1}) - 30f(x_i) + 16f(x_{i-1}) - f(x_{i-2})}{12h^2}$ | $O(h^4)$ |
| **Third Derivative** | |
| $f'''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + 2f(x_{i-1}) - f(x_{i-2})}{2h^3}$ | $O(h^2)$ |
| $f'''(x_i) = \frac{-f(x_{i+3}) + 8f(x_{i+2}) - 13f(x_{i+1}) + 13f(x_{i-1}) - 8f(x_{i-2}) + f(x_{i-3})}{8h^3}$ | $O(h^4)$ |
| **Fourth Derivative** | |
| $f''''(x_i) = \frac{f(x_{i+2}) - 4f(x_{i+1}) + 6f(x_i) - 4f(x_{i-1}) + f(x_{i-2})}{h^4}$ | $O(h^2)$ |
| $f''''(x_i) = \frac{-f(x_{i+3}) + 12f(x_{i+2}) - 39f(x_{i+1}) + 56f(x_i) - 39f(x_{i-1}) + 12f(x_{i-2}) - f(x_{i-3})}{6h^4}$ | $O(h^4)$ |

$$\frac{d^2T}{dx^2} \approx \frac{T_{i-1} - 2T_i + T_{i+1}}{\Delta x^2}$$



$$\frac{d^2T}{dx^2} + h'(T_\infty - T) = 0$$

$$\frac{T_{i-1} - 2T_i + T_{i+1}}{\Delta x^2} + h'(T_\infty - T_i) = 0$$

$$-T_{i-1} + (2 + h'\Delta x^2)T_i - T_{i+1} = h'\Delta x^2 T_\infty$$

$T_0 = T_a \qquad T_n = T_b$

case: $L = 10m \qquad h' = 0.05\ m^{-2}$

$T_\infty = 200K \qquad T_a = 300K, \quad T_b = 400K$

$i = 1$: $-\cancel{T_0} + (2 + h'\Delta x^2)T_1 - T_2 = h'\Delta x^2 T_\infty + T_a$

$i = 2$: $-T_1 + (2 + h'\Delta x^2)T_2 - T_3 = h'\Delta x^2 T_\infty$

$\vdots$

$i = n-1$: $-T_{n-2} + (2 + h'\Delta x^2)T_{n-1} - \cancel{T_n} = h'\Delta x^2 T_\infty + T_b$

plug in numbers and put into matrix form:

$2 + h'\Delta x^2 = 2.2, \quad h'\Delta x^2 T_\infty = 40, \quad n = 5$

$$\begin{bmatrix} 2.2 & -1 & 0 & 0 \\ -1 & 2.2 & 0 & 0 \\ 0 & -1 & 2.2 & -1 \\ 0 & 0 & -1 & 2.2 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} = \begin{bmatrix} 40 + 300 \\ 40 \\ 40 \\ 40 + 400 \end{bmatrix}$$

solve with linear solver.

Since this is tridiagonal, you can use an optimized $O(n)$ solver. Matlab or Julia's `x = A\b` will do this automatically.